

DENNIS KIHUMBA

Full-Stack Web
Developer • REST API
& Backend
Architecture • ICT
Educator

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PROFESSIONAL PROFILE

Full-stack web developer experienced in translating requirements into responsive UI/UX, building modular backend services and REST APIs, integrating relational and NoSQL databases, and implementing authentication, authorization, and security controls. Comfortable across the full application lifecycle — from interface design and data modelling to API integration, deployment, payments, and iterative improvement — with a strong grounding in scalable, SaaS-style system design.

EXPERIENCE

Marjani School

ICT Educator — Web Development & Digital Skills
Current

- Teach practical web development fundamentals — HTML, CSS, and JavaScript — alongside broader digital literacy and problem-solving skills.
- Design hands-on, project-based lessons that guide learners from basic page structure through interactive, functioning websites.
- Translate technical concepts into clear, structured learning materials, reinforcing responsible and effective technology use.

CORE TECHNOLOGY

Frontend

HTML5, CSS3, JavaScript, React, Vite, Tailwind CSS, React Router, responsive & component-based UI development

Backend

Node.js, Express, Python, Flask, modular server architecture, middleware, controllers/services, REST API development

Databases

MongoDB, MySQL, SQLite — data modelling, CRUD operations, relationships, indexing, application integration

Security

Authentication, authorization, password hashing, JWT/session management, RBAC, input validation, CORS, rate limiting, secure configuration

Scalability

Redis caching, load balancing & reverse proxies, stateless services, background jobs/queues, scalable SaaS architecture

Payments

API-based payment integration, transaction workflows, webhook handling, payment status validation, secure server-side processing

Data & Analytics

Python, Pandas, Matplotlib — data cleaning, basic analysis, and visualization

Tooling

Git, GitHub, VS Code, API testing/integration, environment variables, debugging, deployment concepts

SOFTWARE ENGINEERING FOCUS AREAS

- UI/UX & Product Design: translate requirements into user flows, information architecture, responsive layouts, and reusable components; design for accessibility, feedback states, and mobile-first use.
- Backend & API Architecture: design modular Node.js/Express and Python/Flask applications with clear separation of routing, middleware, controllers, services, and data access; build RESTful APIs with consistent status codes, validation, and response structures.
- Database & Data Layer: integrate MongoDB, MySQL, and SQLite; design schemas and relationships; implement CRUD operations, server-side validation, and query-efficient, maintainable data access.
- Auth & Security: implement authentication vs. authorization correctly; apply password hashing, JWT/session flows, RBAC, protected routes, and controls against injection, brute force, and token exposure.
- Scalability & SaaS Design: design stateless services with clear boundaries; apply Redis caching, load balancing/reverse proxies, background job queues, pagination, rate limiting, health checks, and observability as systems grow; structure multi-tenant SaaS systems around users, roles, and subscriptions.

PROFESSIONAL STRENGTHS

- Systems thinking and problem solving
- Requirement analysis and solution design
- Technical communication and presentation
- Leadership and team coordination
- Continuous technical learning